Sri Lankan Black Pepper

Pest and Disease Management

Knowledge Domain: 03 — Pest & Disease Identification, Diagnosis and Control

Project: RAG Chat System for Sri Lankan Black Pepper Knowledge

This document provides comprehensive knowledge on the identification, causes, symptoms, field diagnosis, chemical control, and prevention of all major pests and diseases affecting black pepper (Piper nigrum) cultivation in Sri Lanka, aligned with Department of Export Agriculture (DEA) guidelines.

> ⚠ **Critical Alert:** Pepper Yellow Mottle Virus (PYMV) has NO cure. Quick Wilt (Phytophthora) kills a vine within 2–3 weeks of infection. For both, prevention and early detection are the ONLY effective strategies. Always use certified healthy planting material and maintain drainage and shade control as the first line of defence.

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1. All Pests and Diseases — Quick Reference Summary

This table gives a one-row overview of every major pest and disease. Use it to quickly identify an unknown problem and determine severity before reading the detailed sections below.

Name	Category	Severity	Cause/Spread	Key Symptoms	Primary Control
Pepper Yellow Mottle Virus (PYMV)	Disease — Viral	⚠ CRITICAL	Virus complex; spread by Lace Bug, Mealybug & infected cuttings	Yellow mosaic patches; small irregular leaves; stunted vine; small spikes with half-filled berries; drastic yield decline	NO cure — certified healthy planting material only; destroy infected plants; control insect vectors
Quick Wilt (Phytophthora capsici)	Disease — Fungal	⚠ CRITICAL	Phytophthora capsici fungus thriving in waterlogged, poorly drained soil	Basal stem rots; entire vine wilts and dies within 2–3 weeks of infection	Improve drainage; remove infected parts; Bordeaux mixture or Ridomil MZ 72WP (25g/10L water)
Slow Wilt (Root Decline)	Disease — Complex	● SERIOUS	Root damage from nematodes, mechanical injury, insect feeding, or fungal attack	Yellowing in dry season; recovery after rain; repeated over 1–2 years then permanent decline; 20–30% yield loss	Carbofuran (30g/hole); Gliricidia lopping 4×/year; uproot and burn severe cases; diagnose cause first

Name	Category	Severity	Cause/Spread	Key Symptoms	Primary Control
Vine Borer (Pterolophia annulata)	**Pest — Beetle**	**● SERIOUS**	Longhorn beetle; larvae bore inside vine bark and feed on woody tissue	Whole vine or sections wither; round holes on bark; dead sections do not shed; creamy legless larvae inside	Slash weed; clean 2 ft around vine; remove ground runners; cut & burn infested parts; spray insecticide on surrounding vines
Lace Bug (Diconocoris distanti)	**Pest — Sucking**	**● SERIOUS**	Adults & nymphs suck sap from leaves & spikes; peak in rainy season; primary PYMV vector	Brown spots on leaves; shrivelled immature spikes; few/no berries; damaged spikes fall; severe loss if attack during flowering	Shade & weed control; handpick adults; chemical treatment if severe or at flowering; critical for PYMV prevention
Root Mealybug (Planococcus citri)	**Pest — Sucking**	**● SERIOUS**	White colonies at vine base suck sap from roots & stem below soil; spread by ants	Prolonged yellowing; gradual wilting; whole plant eventually dies; white cottony colonies at soil level	Certified planting material; clean 2 ft around vine; destroy ant colonies; burn infested vines; apply insecticide at base
Thrips	**Pest — Nursery**	**○ MODERATE**	Infected planting material or nursery weeds; primarily a nursery-stage pest	Downward-curled leaf margins; small deformed leaves; stunting, wilting, and death of nursery plants if severe	Healthy cutting material; remove nursery weeds; isolate & burn infected plants; insecticide under extension guidance
Immature Shoot Borer / Leaf Beetles	**Pest — Minor**	**○ MINOR**	Occasional infestation under normal conditions	Shoot tip damage; irregular leaf feeding; limited to small sections of vine	Standard agronomic practices; chemical only if economically significant damage level is reached

> **Severity key:** ⚠ CRITICAL = can kill vine or plantation; immediate action required. ● SERIOUS = significant yield loss if unmanaged. ○ MODERATE / MINOR = manageable with standard practices.

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2. Diseases

2.1 Pepper Yellow Mottle Virus Disease (PYMV)

> **⚠ CRITICAL — NO CURE:** This is the most destructive disease of black pepper in Sri Lanka. Once a vine is infected the virus is systemic throughout the plant — there is no treatment that can reverse or cure it. Prevention and vector control are the only management strategies available.

What Causes It and How It Spreads

- **Pathogen Type:** PYMV is caused not by a single virus but by a complex of viruses — this is why it cannot be treated with any fungicide or antibiotic.
- **Insect Vectors:** The primary insect vectors (carriers) are the Pepper Lace Bug (Diconocoris distanti) and Mealybugs. When these insects feed on an infected vine, they acquire the virus and carry it to the next healthy vine they feed on. This is why lace bug and mealybug control is central to PYMV management.
- **Infected Cuttings:** Infected planting material is the second major spread route — cuttings taken from a PYMV-positive mother vine carry the virus and introduce it directly into the nursery or new field, sometimes before symptoms even appear.
- **Systemic Spread:** Once a vine is infected, the virus spreads systemically through all plant tissue. The entire vine is a virus source. Partial removal is not sufficient — the whole plant must be destroyed.
- **What Does NOT Spread It:** PYMV is not spread by soil or water — only by insect feeding on infected plant tissue, and by using infected planting material. This makes planting material hygiene and vector control the two pillars of prevention.

Symptoms — What to Look For

- **First Sign (early detection):** The earliest sign is yellowish spots appearing on young, newly emerging leaves. Older leaves may look relatively normal at this early stage — check new growth first.
- **Leaf Symptoms:** As the disease progresses, leaves become small, irregular in shape, and develop yellow mosaic patches — a characteristic patchwork of yellow and green areas that does not wash off or improve with irrigation or fertilisation.
- **Vine Structure:** Short internodes — the vine fails to elongate at a normal rate, and the gaps between leaf nodes become noticeably shorter.
- **Yield Impact:** Spikes become small and poorly formed; berries are few, undersized, and half-filled — yield drops dramatically even before the vine shows obvious dying symptoms.
- **Progressive Decline:** Gradual and drastic yield decline occurs over successive seasons. A PYMV-infected vine left in the field not only produces little — it continuously spreads the virus to neighbours through insect vectors.

Control Measures

1. Use only certified, healthy planting material from disease-free mother vines that have been inspected and confirmed PYMV-free. This is the single most important preventive action in the entire management programme.
2. Never take cuttings from any vine showing yellowing, mosaic leaf patterns, stunted growth, or poorly filled spikes — even mild early symptoms indicate active infection.
3. Immediately destroy infected vines — uproot the entire plant, roots and all, and burn it on-site. Do not compost. Do not move infected plant material across the field. Do not leave debris.
4. Control lace bug and mealybug populations across the entire plantation at all times — vector control is the primary mechanism for slowing PYMV spread from plant to plant.
5. Inspect all vines regularly — at minimum monthly — focusing on young leaves for early mosaic signs. Early detection before the virus has spread to neighbouring vines significantly limits plantation-level impact.
6. Immediately isolate any suspicious vine — stop cutting or pruning operations near it until it is confirmed infected or clear; pruning tools can carry sap with virus particles between plants.

> **No chemical cure:** There is no fungicide, insecticide, or any treatment that cures PYMV once a vine is infected. Every day an infected vine remains in the field it is spreading virus to neighbouring vines via lace bugs and mealybugs. Prompt destruction protects the rest of your plantation.

2.2 Quick Wilt — Phytophthora Root and Stem Rot

> **⚠ CRITICAL:** An entire vine can die within 2–3 weeks of infection. Quick Wilt acts rapidly — especially in wet or waterlogged conditions. Correct drainage is the most important long-term control. Chemical response must happen within 24–48 hours of detecting basal rot.

What Causes It and How It Spreads

- **Pathogen:** Caused by the soil-borne water mould Phytophthora capsici — a fungal-like organism that thrives in waterlogged, poorly drained soil. It is not a true fungus but behaves like one and responds to certain fungicides.
- **Infection Route:** The pathogen infects the base (basal part) of the vine stem first, where it enters through natural openings or small wounds in the bark. It then spreads downward into the root system, progressively destroying the plant's ability to take up water and nutrients.
- **Risk Factors:** Waterlogged or poorly drained soil is the primary enabling condition — Phytophthora capsici cannot spread aggressively in well-drained soils. Flooding events, areas with poor drainage infrastructure, and excessive Gliricidia shade keeping soil wet all significantly raise the risk.
- **Soil Persistence:** The pathogen can remain dormant in the soil for years and reactivate whenever drainage conditions worsen — making permanent drainage improvement a non-negotiable long-term solution, not just a one-time fix.

Symptoms — What to Look For

- **First Sign:** The base of the vine appears dark, water-soaked, and soft when pressed — this is the earliest visible sign, often appearing before wilting of the upper vine begins.
- **Diagnostic Test:** When the basal stem tissue is cut open it appears brown and rotten — compared to healthy, firm, white tissue in an uninfected vine. This cross-section check is a reliable field test.
- **Root Damage:** Root rot spreads outward and downward from the stem base — as more roots are destroyed, the vine progressively loses its ability to absorb water.
- **Sudden Wilt:** The entire vine wilts rapidly once the root system is sufficiently damaged — all leaves collapse together, distinguishing it from gradual wilts caused by other conditions.
- **Timeline:** Death of the entire vine typically occurs within 2 to 3 weeks from first visible symptoms. No recovery is possible once basal rot is advanced.

Control Measures

7. Improve drainage immediately and permanently around affected vines and across the whole plantation. Create drainage channels. Raise planting beds in flood-prone areas. This is the most important long-term action.
8. Regulate Gliricidia shade — dense, unlopped Gliricidia keeps soil moist and creates conditions Phytophthora capsici thrives in. Maintain the recommended 4–5 m canopy height.
9. At the first sign of basal rot: remove and burn all infected plant material — rotten stem sections, roots, and surrounding soil debris. Do not leave any infected material in the field.
10. Apply Ridomil MZ 72WP (Mancozeb 64% + Metalaxyl 8%) at 25 g dissolved in 10 litres of water — spray or drench directly at the base of the infected vine and all immediately surrounding healthy vines.
11. Apply Bordeaux mixture as a preventive spray at the base of all vines in flood-prone or poorly drained areas at the start of each wet season (Maha and Yala).
12. Do not replant pepper in areas with a confirmed history of Quick Wilt until drainage infrastructure is fully corrected and an adequate fallow period has passed.

2.3 Slow Wilt — Root Decline Syndrome

> ● **SERIOUS:** Causes 20–30% yield decline over 1–2 seasons. Develops slowly and is easy to overlook until significant damage is done. Unlike Quick Wilt, it has multiple possible causes — accurate diagnosis of the actual cause is essential before applying any chemical treatment.

What Causes It and How It Spreads

- **Complex Cause:** Slow Wilt is not a single-pathogen disease — it is a root decline syndrome triggered by multiple possible causes acting alone or together.
- **Nematodes (Primary):** Root-knot nematodes are the most common cause in Sri Lanka. These microscopic soil worms invade roots and create galls (knots) that block water and nutrient movement. Affected roots cannot be fully restored — prevention is far more effective than treatment.
- **Mechanical Damage:** Mechanical root damage from careless cultivation, heavy earthworm activity, or soil disturbance near the vine base opens entry points for secondary fungal infections that accelerate the decline.

- **Other Causes:** Insect root feeders and fungal root pathogens can independently cause or significantly worsen slow wilt — applying a nematicide when a fungus is the actual cause will have no effect.

Symptoms — What to Look For

- **Seasonal Yellowing Pattern:** The hallmark sign is seasonal yellowing — leaves turn yellow during the dry season (drought) but return to green after monsoon rains arrive. This cycle repeats for one to two seasons before decline becomes permanent.
- **Progressive Worsening:** Recovery after rain becomes progressively less complete over time — the vine appears greener after rain but never returns to full vigour. This gradual worsening is distinctive and should trigger investigation.
- **End Stage:** In the final stage the vine becomes permanently yellow and fails to recover even when adequate rainfall is present — at this point root damage is too severe to reverse.
- **Yield Loss:** Approximately 20–30% yield decline occurs in affected plantations, often before vines show obvious dying symptoms — making early detection through the seasonal pattern critical.

Control Measures

13. Confirm the actual cause first — nematodes, fungal pathogen, mechanical damage, or insect damage — before applying any chemical. Root and soil examination, or extension officer support, is required for accurate diagnosis.
14. Prevention is far more effective than treatment: add Carbofuran (3 g per nursery pot at potting stage; 30 g per planting hole at field planting) to protect roots from nematodes from the very beginning.
15. Maintain Gliricidia lopping at four times per year — DEA experimental evidence confirms that this level of lopping significantly reduces nematode populations in the root zone over time.
16. Maintain high soil organic matter through Gliricidia lopped material and compost application — beneficial organisms in organic-rich soils naturally suppress nematode populations.
17. Implement soil conservation practices — avoid erosion, minimise soil disturbance around vine roots, and ensure the root zone is not mechanically damaged during weeding or cultivation.
18. If serious root damage is found and the vine has not recovered after two seasons of correct management, uproot and destroy the affected vine to prevent it becoming a nematode reservoir for neighbouring plants.

> **Do not treat blindly:** Applying a nematicide when the actual cause is a fungus, or vice versa, wastes resources and delays effective management. Correct diagnosis saves time and money. Contact your nearest DEA extension officer for root inspection support.

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3. Pests

3.1 Vine Borer — Pterolophia annulata

> **Emerging Threat:** Vine borer damage has been reported recently in Sri Lanka and is considered a rising serious pest. Farmers and extension officers should remain alert to its spread and report new incidences to the DEA for monitoring.

Biology and Behaviour

- **Pest Type:** Vine borer belongs to the longhorn beetle family (Cerambycidae). The adult beetle lays eggs on or near the surface of pepper vine bark.
- **Damage Mechanism:** Larvae hatch and bore into the bark and woody tissue of the vine, feeding from inside the stem. Because all damage is internal, infestations are often not detected until the vine section or entire vine begins to yellow and wither.
- **Scope of Damage:** Damage can affect either a section of the vine (where larvae are feeding) or the entire plant depending on where and how many larvae are present. Damage to the main stem is most severe.
- **Entry Points:** The presence of ground runners (stems rooted along the soil surface) and low lateral branches touching the ground are believed to facilitate adult beetle access and egg laying — making their removal a key prevention step.

Symptoms — What to Look For

- **Primary Sign:** The whole vine or specific sections yellowing and withering — may initially be mistaken for drought stress, root disease, or fertiliser deficiency.
- **Key Diagnostic Sign:** Small, round holes on the bark surface — these are entry or emergence holes created by larvae burrowing in or adult beetles emerging out. This is the most distinctive and diagnostic field sign.
- **Distinguishing Feature:** Dead sections of the vine remain firmly attached to the support and do not shed or fall — unlike natural leaf senescence where dead tissue eventually drops. This is a distinguishing feature from other causes of vine die-back.
- **How to Confirm:** Confirmation test: break open the dead or dying section of vine — creamy-coloured, legless larvae will be visible feeding inside the woody stem tissue.

Control Measures

19. Slash weed the entire field — remove all tall weeds that harbour adult beetles and provide egg-laying habitat near vines.
20. Clean weed within 2 feet (approximately 60 cm) of each vine base — a clear, weed-free zone around the vine base removes direct beetle access routes.
21. Cut and remove all ground runners and all lower plagiotropic (lateral) branches that are touching or lying on the ground — these are the most vulnerable entry points for egg-laying adult beetles and must be maintained removed year-round.
22. Where bore holes or infested sections are confirmed: cut out the affected section or remove the entire plant if damage is widespread. This is not optional — larvae left in situ will continue to kill the vine and potentially spread to nearby plants.
23. All removed and infested vine parts must be destroyed immediately by burning — never leave debris on the ground or in the field. Unburned material allows larvae to complete their life cycle and adult beetles to emerge.
24. After burning, apply a recommended insecticide to the vines immediately surrounding the removed plant as a protective treatment to prevent adult beetles relocating.

3.2 Lace Bug — Diconocoris distanti

> **Double Threat:** Lace Bug causes direct yield loss through spike and leaf damage AND is the primary insect vector for Pepper Yellow Mottle Virus (PYMV). Controlling lace bug is therefore essential both to protect current yield and to prevent the spread of an incurable viral disease.

Biology and Behaviour

- **Feeding Habit:** Both adult and nymph lace bugs cause damage by piercing plant cell walls and sucking cell sap from leaves and developing spikes. Nymphs and adults feed together on the same plant tissue.
- **Seasonal Pattern:** Lace bugs are present on pepper plantations throughout the year but populations increase rapidly during and after the rainy season when humidity is high and shade is dense — both conditions created by unlopped Gliricidia.
- **PYMV Vector Mechanism:** When a lace bug feeds on a PYMV-infected vine it acquires the virus. When it then moves to a healthy vine and feeds again, it injects the virus into the healthy plant. This is called persistent vector transmission and is the primary route by which PYMV spreads within a plantation.

Symptoms — What to Look For

- **Leaf Damage:** Brown spots appear on leaves in the pattern of concentrated feeding — multiple tiny puncture marks create a speckled brown appearance. This is distinct from fungal leaf spots which often have a defined border.
- **Spike Damage:** Immature spikes become shrivelled, brown, and produce few or no berries — the feeding damage during early spike development prevents normal berry formation and filling.
- **Spike Shedding:** Damaged spikes dry completely and fall from the vine to the ground — visible fallen spikes are a field indicator of active or recent lace bug damage.
- **Critical Timing — Flowering Stage:** If lace bug attack coincides with the flowering stage of spike development, the yield loss is severe and irreversible — berries that fail to set at flowering cannot be recovered in that season. Timing of treatment relative to the flowering stage is critical.

Control Measures

25. Shade control — lop Gliricidia regularly to maintain the recommended 4–5 m canopy height. Dense shade from unmanaged Gliricidia is the single biggest driver of lace bug population increases. This is the most important non-chemical control measure.
26. Weed control — remove all weeds from the plantation, especially within 2 feet of vine bases. Weeds serve as alternative host plants and harborage sites for lace bug adults and nymphs.
27. Handpick adult lace bugs during regular vine inspections and destroy them. Handpicking is effective for early-stage or low-population infestations and keeps the pest below economic damage levels without chemicals.
28. If the infestation is severe OR if the attack is occurring or approaching the flowering stage, apply a recommended insecticide immediately. Do not delay treatment waiting to see if the population drops naturally — spike damage at flowering is permanent.

1. Treat lace bug as a PYMV prevention priority, not just a yield pest. Even at low population levels, lace bugs on an undetected infected vine can spread PYMV widely. Proactive management at all times is safer than reactive treatment.

3.3 Root Mealybug — Planococcus citri

Biology and Behaviour

- **Appearance:** Root mealybugs are small, oval, white or grey insects covered in a waxy powder. They live in dense colonies at the base of the pepper vine and just below the soil surface — entirely hidden from view unless you scrape back the topsoil.
- **Feeding Location:** They feed by sucking cell sap from the roots and the main stem below ground level. Because all feeding is underground the vine shows stress symptoms above ground (yellowing, wilting) before any pest is seen.
- **Ant Association — Key Indicator:** Ants have a mutualistic relationship with root mealybugs — ants protect the mealybug colonies from natural predators and actively carry mealybugs to new plants in exchange for the sweet honeydew secretions mealybugs produce. Active ant trails at vine bases are the most reliable above-ground early indicator of mealybug infestation.
- **PYMV Risk:** Root mealybug is also classified as a secondary PYMV vector — it can acquire and transmit the virus when feeding on infected vines, adding further reason to control it proactively.

Symptoms — What to Look For

- **Yellowing:** Prolonged yellowing of leaves that develops over weeks to months — unlike Quick Wilt which collapses suddenly, mealybug damage creates a slow, progressive yellowing.
- **Wilting:** Progressive wilting of the vine despite adequate rainfall or irrigation — the root damage impairs water uptake even when water is available in the soil.
- **Ant Activity — Act on This Sign:** Active ant trails running to and from the vine base are often the first sign visible above ground. If you see regular ant activity at a vine base, immediately scrape back the topsoil and mulch to check for mealybugs.
- **Underground Sign:** White cottony masses (resembling small balls of cotton wool) visible at or just below the soil surface around the vine base when mulch or soil is scraped back.
- **Outcome if Untreated:** If left untreated, the entire vine dies — mealybug infestations do not resolve on their own and worsen progressively.

Control Measures

2. Use only certified, healthy planting material — root mealybug can be introduced into a new field through contaminated nursery plants without any visible symptoms at the time of planting.
3. Maintain vine vigour through the correct fertiliser schedule — a vigorously growing vine tolerates low levels of mealybug pressure better and resists the associated PYMV risk more effectively than a weak, stressed vine.
4. Clean weed within 2 feet of each vine base. Slash remaining field weeds. Weeds serve as alternative mealybug host plants and maintain ant colonies that protect and spread the pest.

5. Actively destroy ant colonies around vine bases — this is the step most commonly missed. Without ant protection from predators, mealybug colonies weaken naturally. Apply ant bait or physical disruption of nests systematically across the plantation.
6. Conduct monthly inspections — scrape back soil and mulch near vine bases to detect white cottony colonies before the infestation is severe enough to cause obvious vine symptoms.
7. Destroy heavily infested vines that are clearly beyond recovery — do not leave them to be a permanent source of infestation for neighbouring plants and continued ant-mediated spread.
8. Apply a recommended contact insecticide to the vine base and root zone with DEA extension officer guidance for correct product selection, rate, and method.

3.4 Thrips — Nursery-Stage Pest

> **Nursery Alert:** Thrips are primarily a nursery pest. A severe infestation can destroy entire batches of cuttings before they ever reach the field — representing a total loss of several months of nursery work. Strict hygiene in the nursery at all stages prevents field-scale impact.

Why the Nursery Stage is Critical

Pepper plants in the nursery are in a confined, high-humidity environment — ideal conditions for thrips populations to establish and spread rapidly from one cutting to another. A single infected batch of planting material introduced into an otherwise clean nursery can trigger plantation-wide crop loss if not caught early.

Symptoms — What to Look For

- **First Sign:** Downward curling of leaf margins — one of the earliest and most consistent signs in nursery plants. Check the underside of leaves and the growing tips where thrips prefer to feed.
- **Leaf Deformation:** Leaves are unusually small and distinctly deformed in shape — different from the typical leaf distortion caused by other problems such as mineral deficiency.
- **Severe Cases:** Severe infestation causes stunting of overall plant growth, wilting, and eventual death of individual nursery plants — and rapid spread to neighbouring plants in the nursery bed.

Control Measures

9. Select only healthy, insect-free planting material from clean, confirmed disease-free mother vines for all propagation. Never reuse cuttings from any nursery plant that showed symptoms.
10. Keep nursery beds completely weed-free at all times — weeds are the primary reservoir from which thrips populations establish and re-establish.
11. Separate and physically isolate any nursery plant showing leaf curling or deformation immediately — before confirming it is thrips, isolation prevents spread.
12. Remove and burn all infected leaves and affected plants from the nursery. Do not compost.
13. If the infestation has spread widely within the nursery, burn the entire affected batch immediately and thoroughly clean and disinfect the nursery structure and beds before starting a new batch with fresh, certified material.
14. Apply a recommended insecticide to nursery plants under the guidance of a DEA extension officer when chemical control is necessary.

3.5 Minor Pests — Immature Shoot Borer and Leaf-Eating Beetles

Immature shoot borers and leaf-eating beetles are classified as minor pests because under normal cultivation conditions their damage levels remain below the threshold of economic significance. Damage is localised — typically to shoot tips or irregular patches of leaf tissue — and does not progress to vine death or plantation-scale yield loss.

Standard agronomic practices — maintained shade control, regular weeding, correct fertilisation for vine vigour, and clean cultural practices — are sufficient to keep these pests at tolerable levels. Chemical control is not routinely recommended. It should only be considered if population levels increase significantly and visible damage becomes widespread across multiple vines. In such cases, consult a DEA extension officer before applying any insecticide.

4. Field Diagnostic Guide — 'What Do I See?'

Use this table when a farmer describes what they are seeing in the field and needs immediate identification and action guidance. Match the visible symptoms to the most likely cause and follow the action steps.

What the Farmer Sees	Most Likely Cause	Recommended Action	Urgency
Yellow mosaic patches on young leaves; small irregular leaves; short internodes; vine stunted; small spikes with half-filled berries	**Pepper Yellow Mottle Virus (PYMV)**	Mark and destroy the infected vine immediately by uprooting and burning — do NOT compost. Never take cuttings from this vine. Check all nearby vines. Control lace bug and mealybug to stop further spread.	**Destroy today**
Vine wilts suddenly and completely; base of stem is dark, soft, or rotten when pressed; entire vine dies within 2–3 weeks	**Quick Wilt (Phytophthora)**	Check and improve drainage around the vine immediately. Remove all infected plant parts and burn them. Apply Ridomil MZ 72WP (25g in 10L water) or Bordeaux mixture to the base of infected vine and surrounding healthy vines.	**Act within 24–48 hrs**
Vine leaves yellow during dry season but recover to green after monsoon rains; this repeats for 1–2 seasons before vine gradually permanently declines	**Slow Wilt (Root Decline)**	Inspect roots for nematode galls or fungal rot. Apply Carbofuran (30g/planting hole). Lop Gliricidia 4×/year. Maintain soil organic matter. Uproot and destroy if no recovery after 2 seasons. Confirm cause before chemical treatment.	**Act within the week**
Whole vine or sections wither and yellow; small round holes visible on bark surface; dead	**Vine Borer (Pterolophia annulata)**	Break open the dead section — confirm creamy legless larvae inside. Slash all field weeds. Clean 2 ft around vine base. Remove all	**Act within the week**

What the Farmer Sees	Most Likely Cause	Recommended Action	Urgency
sections remain attached (do not shed/fall off vine)		ground runners and low lateral branches touching soil. Cut and burn infested parts. Spray insecticide on surrounding healthy vines.	
Brown spots on leaves; immature spikes shrivelled and producing few or no berries; damaged spikes dried and fallen to ground	**Lace Bug (Diconocoris distanti)**	Lop Gliricidia immediately to reduce shade. Weed 2 ft around vine. Handpick and destroy adult lace bugs. If at flowering stage or severe: apply insecticide immediately. Also treat to reduce PYMV risk since lace bugs are primary virus vectors.	**Immediate if flowering**
Vine gradually yellowing and wilting despite adequate rain; white cottony clusters visible at soil level around vine base; ants actively moving around base	**Root Mealybug (Planococcus citri)**	Clean weed 2 ft around vine. Destroy ant colonies — ants carry mealybugs from plant to plant. Scrape back soil at vine base to confirm white colonies. Apply insecticide at base under extension guidance. Burn heavily infested vines.	**Act within the week**
Nursery plants only — leaf margins curling downward; leaves small and deformed; plants stunting, wilting, or dying in batches	**Thrips**	Remove and burn infected nursery plants immediately. Clear all weeds from nursery beds. Isolate any suspicious plants. Assess and if necessary replace planting material source. Apply insecticide under extension guidance.	**Act immediately**
Spikes have brown spots but vine is otherwise healthy; lace bug adults visible on leaves; wet season underway	**Lace Bug — early stage**	Implement shade control (lop Gliricidia) and weeding first. Monitor population. Handpick adults. If damage is progressing toward flowering stage — treat chemically immediately. Early treatment preserves spike yield.	**Monitor & treat if increasing**

> **How to use this table:** Read across the row matching what you see in the field. The 'Urgency' column tells you how quickly to act. For Critical or immediate cases, start action the same day — do not wait for further symptoms to develop.

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5. Chemical Control Reference

Specific chemical control options approved and recommended for black pepper pest and disease management in Sri Lanka. Always follow label instructions and consult a DEA extension officer before applying, especially for products not previously used on your plantation.

Disease / Pest	Chemical Name	Active Ingredient	Rate / Dose	Application Method	When to Apply
Quick Wilt (Phytophthora capsici)	**Ridomil MZ 72WP**	Mancozeb 64% + Metalaxyl 8%	25 g dissolved in 10 L of water	Spray or drench at the base of infected vine and surrounding healthy vines	At first sign of basal rot; after removing infected plant material
Quick Wilt — Preventive	**Bordeaux Mixture**	Copper sulphate + Lime (1% solution)	Standard 1% Bordeaux preparation	Spray to basal stem and root zone; especially after heavy rain or flooding events	Preventive at the start of the wet season in high-risk, poorly drained areas
Slow Wilt — Nematode Prevention	**Carbofuran (granular)**	Carbofuran	3 g per nursery pot; 30 g per planting hole at field establishment	Incorporate into nursery growing medium or planting hole soil — not a foliar spray	At nursery potting stage OR at time of field planting only
Lace Bug / Root Mealybug / Vine Borer	**Recommended insecticide (seek DEA guidance)**	As advised by DEA extension officer for specific pest	As per product label	Spray on affected vines and immediately surrounding healthy vines	When pest population is high; when damage occurs at or near the flowering stage
Thrips (nursery stage)	**Recommended insecticide (seek DEA guidance)**	As advised by DEA extension officer	As per product label	Spray on affected nursery plants; treat nursery beds	After isolating and removing infected plants; when chemical control is necessary

Important: Chemical treatment should always be the final layer of defence — after improving drainage, removing infected material, controlling shade and weeds, and using certified healthy

> planting material. Over-reliance on chemicals without correcting the underlying agronomic conditions leads to disease recurrence and potential pesticide resistance.

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6. Integrated Pest and Disease Prevention

Prevention is significantly more effective and less costly than treatment for all major black pepper pests and diseases in Sri Lanka. The following checklist consolidates every preventive action a farmer should take, what it protects against, and when to do it.

Preventive Action	Protects Against	Timing
Use only certified, healthy planting material from disease-free, inspected mother vines — NEVER use cuttings from a vine showing yellowing, mosaic, or stunting	**PYMV, Thrips, Root Mealybug**	Before nursery establishment
Add Carbofuran (3 g) to each nursery pot growing medium at potting stage to protect roots from nematodes from day one	**Slow Wilt (nematodes)**	At nursery potting
Add Carbofuran (30 g) into each planting hole before transplanting pepper seedling	**Slow Wilt (nematodes)**	At field planting
Establish Gliricidia support trees at least 6 months before planting pepper — good shade from the start reduces lace bug habitat	**Lace Bug, PYMV spread**	6 months before planting
Ensure proper field drainage — soil must never remain waterlogged for more than 48 hours around vine bases; create drainage channels in low-lying areas	**Quick Wilt (Phytophthora)**	Year-round; critical in wet season
Lop Gliricidia support trees 3–4 times per year (4× per year in wet zone) to control shade and suppress lace bug habitat	**Lace Bug, PYMV, Slow Wilt**	Every quarter; 4× in wet zone
Clean weed within 2 feet (≈60 cm) of each vine base; slash remaining field weeds — removes pest harborage and ant pathways for mealybugs	**Root Mealybug, Vine Borer, Lace Bug**	Year-round
Remove all ground runners and lower lateral (plagiotropic) branches touching the soil — primary egg-laying sites for vine borer adults	**Vine Borer**	Year-round; especially dry season
Inspect vine bases monthly — scrape back soil and mulch to look for white cottony mealybug colonies and active ant trails	**Root Mealybug**	Monthly
Inspect bark surface monthly for small round holes — early detection of vine borer before larvae kill the vine section	**Vine Borer**	Monthly
Destroy infected plants immediately by uprooting and burning — never compost or leave in field;	**PYMV, Quick Wilt, Vine Borer, Thrips**	Whenever detected

Preventive Action	Protects Against	Timing
one infected plant left can spread disease to many neighbours		
Apply Bordeaux mixture preventively at the base of vines in poorly drained or flood-risk areas at the start of the wet season	**Quick Wilt (Phytophthora)**	Start of Maha and Yala wet seasons
Maintain vine vigour through correct fertilizer schedule — a well-nourished vine has significantly greater natural resistance to all pest and disease pressure	**All pests and diseases**	Maha & Yala seasons
Destroy ant colonies near vine bases — ants protect and actively transport mealybug colonies to new plants; controlling ants breaks the mealybug cycle	**Root Mealybug**	Whenever ant activity detected
Keep nursery environment strictly clean — weed-free, with no infected plants allowed near healthy cuttings	**Thrips, PYMV (via infected cuttings)**	Throughout nursery period

> **Core principle:** A well-managed plantation — with correct drainage, proper shade, clean weeding, vigorous vines from correct fertilisation, and certified planting material — has dramatically lower pest and disease incidence than a neglected one. Most major pest and disease outbreaks in Sri Lankan black pepper are preventable with good agronomic practice. Chemical treatment treats the symptom; good management treats the cause.